

Physics of Cognition: Homework 5

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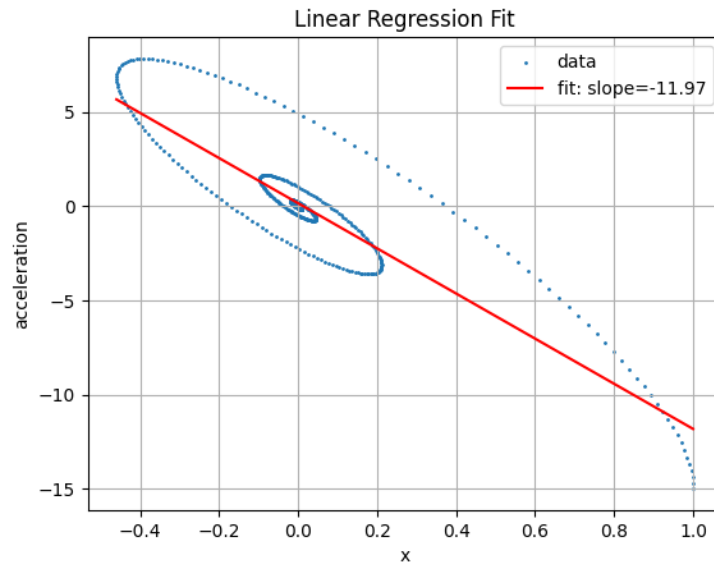
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1.1: Leaky spring

1

Yes, the fit succeeds perfectly. The predicted k/m is only 0.03 from the actual value.

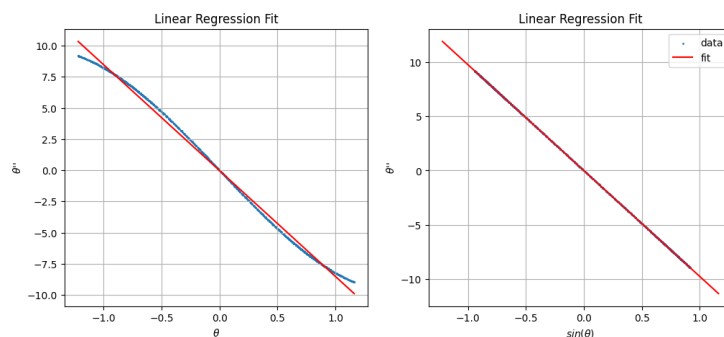
2



The linear model doesn't work for this data. It can approximate one degree of freedom but fails to generalize when there are two (or multiple) since there aren't enough variables.

The velocity was information not part of the regression but was used as a basis when simulating the spring. If we concatenate the velocity in the regression's x , we could get a good predictor.

1.2: Pendulum



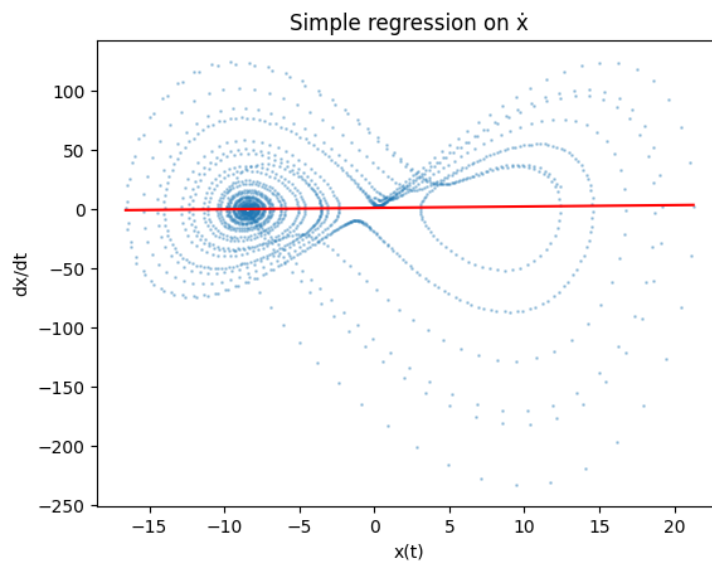
Actual g/L : 9.8

Inferred g/L : 9.7804

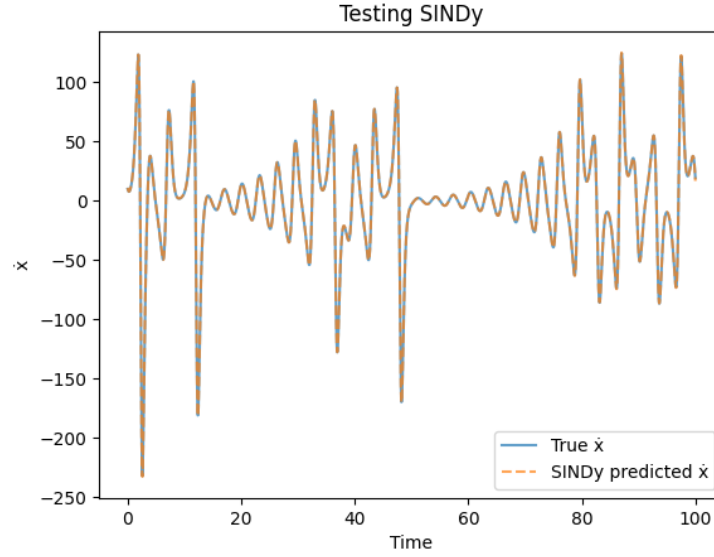
It's hard to get a linear approximation of $\sin(\theta)$ which is fundamentally a nonlinear function. This is why the first model wasn't as accurate as the second which included $\sin$ in its basis.

1.3: Lorenz attractor and SINDy

Bad regression



Good regression using Lasso (this is graphed with time-series).



Let's look at the coefficients it gave:

Predicted:

$$dx/dt = -11.64*x + 10.41*y + 0.01*x^2 + 0.05*x \ z$$

$$dy/dt = 27.11*x + -1.44*y + -0.48*z + -0.10*x^2 + 0.07*x \ y + \dots$$

$$dz/dt = -0.14*x + -0.02*y + -3.21*z + 0.07*x^2 + 1.04*x \ y + \dots$$

Actual:

$$dx/dt = \text{sig}*(y - x) \rightarrow -10*x + 10*y$$

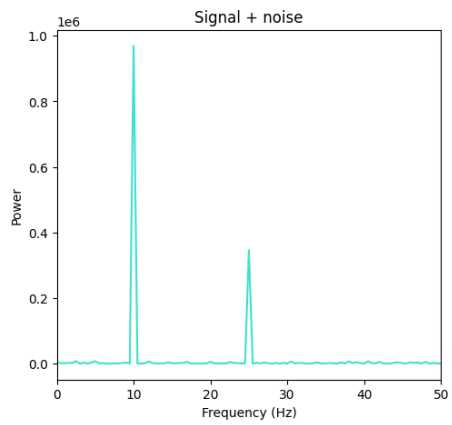
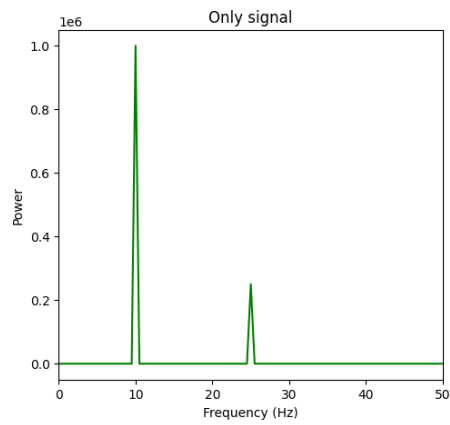
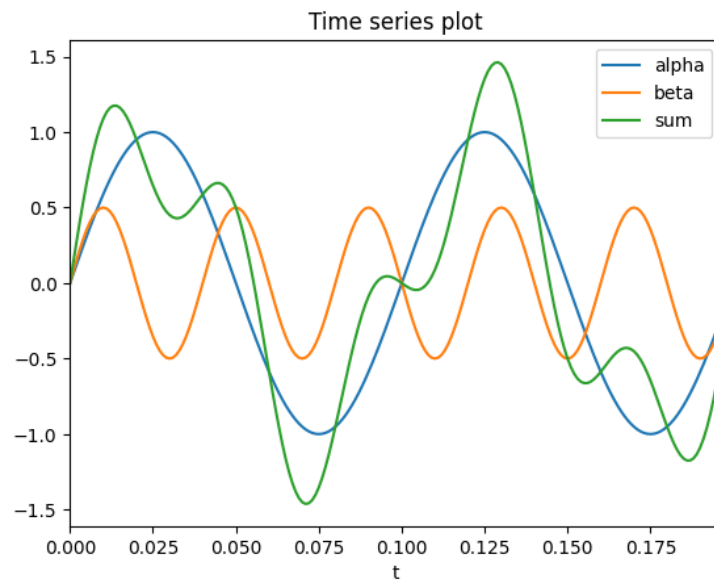
$$dy/dt = x(r - z) - y \rightarrow 28*x - y - x*z$$

$$dz/dt = xy - \text{beta}*z \rightarrow x*y - 2.67*z$$

They look pretty good! Lasso regression is able to find the weights for each term (defined by a library) such that we can predict the rate at each variable. Spread across many timesteps, we have an overdetermined system that can be solved for the best weights. To minimize noise and overfitting, Lasso penalizes non-zero weights to create a sparse model of the data. And for the Lorenz attractor, this works really well.

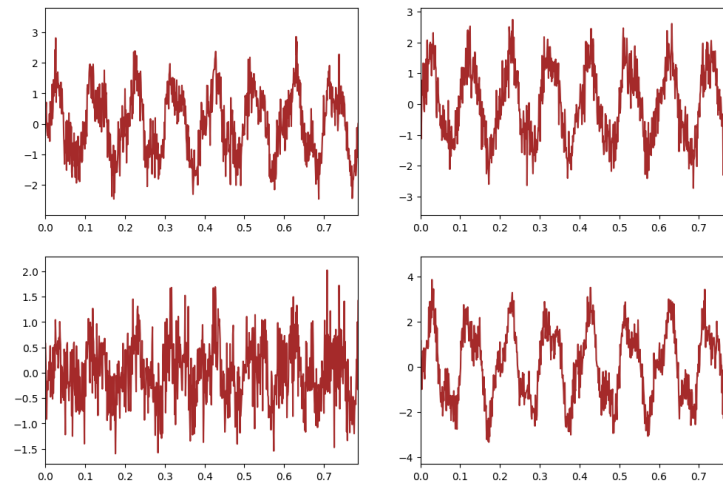
2.1: Single electrode

The frequencies are at 10 Hz and 25 Hz as expected. Even with the Gaussian noise, the signals remain easily identifiable with the frequency plot.

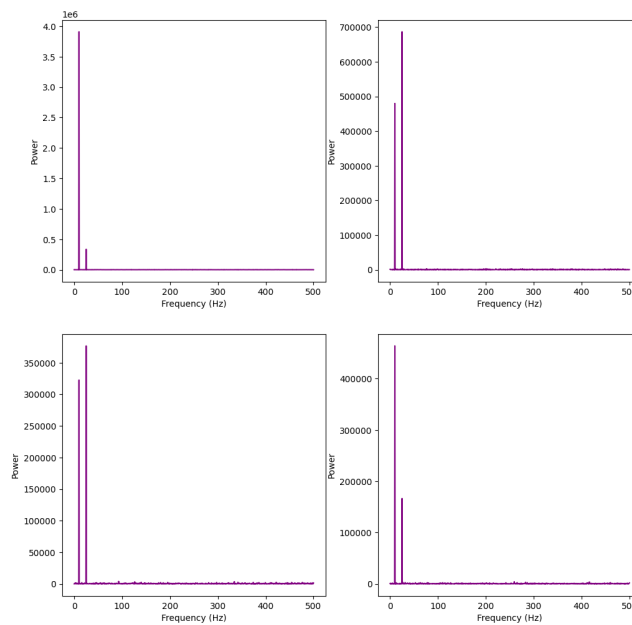


2.2: Mixed signals

Time-series plot



FFT Power Spectrum



All the frequency peaks occur at the same x location meaning that despite mixing, each electrode can still capture the same underlying signal with more or less background noise. The peak frequency heights vary depending on the relative signal strengths (we sometimes see massive differences in magnitude between each plot). This reveals that with a similar setup in the brain, we could obtain underlying rhythms even if the sensitivity differs with each source.

2.3: Better model

We would want something like an exponential function that decays as the distance between the signal and electrode increases. This is more realistic since not every electrode is as sensitive to the signal as every other (there is a spatial dimension embedded into the weights).

3: Reflection

1. Similar to how we can extract the coefficients of a pendulum using a bunch of simulated datapoints, the brain could have a deep dynamical system containing its own equations with the interaction of each neuron. Given the dynamics, there are certain stable frequencies we can extract with FFT—in our exercises this was simply a sum of two sine curves, but FFT can yield useful results even if the source is more complex.
2. Prior to this paper, the neuron was thought to follow the "integrate-and-fire" model where they temporally integrate independent potentials (EPSPs) and fire after a threshold. This should result in a regular output since many independent samples should approximate a normal curve due to Central Limit Theorem, yet the monkey data contradicts with highly irregular firings. Such results overturned the previous paradigm and led us to believe that neurons operate more like millisecond-resolution coincidence detectors, which is plausible if there are strong nonlinearities in the dendrites. Fundamentally, we started to believe that information wasn't purely carried in the average firing rate, but the exact timing and synchronization held a huge part, known as temporal coding. This form of understanding is still seen today with concepts like E/I balance.
3. This homework was quicker, taking 4 hrs. I enjoyed the progression from the spring to pendulum to more chaotic systems and taking a look at FFT which is a concept I've only briefly come across. With this homework especially, I feel like I fully understand the code and simulations we do and how that ties in with what we're learning.

Code